

CHAPTER ONE

INTRODUCTION

1.0 Introduction

The present study develops and evaluates a web-based system for predicting the Big Five personality traits from social media text. The system is implemented as a Django application and combines text preparation, post selection, contextual embedding extraction, data augmentation and machine learning models. The traits considered are Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism, commonly represented as OCEAN.

The study is motivated by the increasing availability of naturally occurring text on online platforms. Written comments and posts may contain linguistic patterns that are related to stable personality differences. At the same time, personality prediction from text remains difficult because social media content is varied, noisy and often limited for individual users. A useful system must therefore select suitable text, represent its meaning adequately and report predictions with appropriate caution.

The implemented system uses the PANDORA personality dataset for the main experiment. The dataset contains English-language online comments linked with Big Five personality scores. The data are organized into separate training, validation and test sets so that model development and final evaluation are separated. This gives the study a clearer experimental basis than relying only on a small collection of local records.

The machine learning pipeline compares two prediction approaches. The first uses Lasso or ElasticNet regression on pooled BERT embeddings. The second uses a stacked bidirectional Long Short-Term Memory (LSTM) model to learn from sequences of comment embeddings. In both approaches, a baseline selection method is compared with a Q-learning selection method, while augmentation is either included or omitted. This creates eight experimental conditions and makes it possible to examine the contribution of model type, text selection and augmentation separately.

BERT is used to transform selected comments into contextual numerical representations. These representations are supplied either to the sparse regression model or to the sequence model. The augmentation component generates synthetic embedding and

personality-score pairs from the training data only. The final results are recorded with the selected model condition, data split, decision thresholds and performance measures. The system therefore provides a practical environment for examining personality prediction rather than treating the task as a purely theoretical exercise.

1.1 Background and Motivation to the Study

Personality describes relatively consistent patterns in thought, emotion and behaviour. The Big Five model is one of the most widely used frameworks for describing these patterns. It groups personality into Openness to Experience, Conscientiousness, Extraversion, Agreeableness and Neuroticism. These traits have been associated with outcomes in education, work, wellbeing and interpersonal relationships (Goldberg, 1993; McCrae & Costa, 2008).

Traditional personality assessment commonly relies on questionnaires such as the Big Five Inventory. Questionnaire results remain important because they provide a direct self-report measure of personality. However, self-report data may be affected by social desirability, incomplete recall and the respondent's interpretation of the questions. Questionnaire administration can also be difficult to repeat frequently for large groups of people (Paulhus & Vazire, 2007).

Social media and online discussion platforms provide another source of behavioural evidence. Users leave text through posts, comments, replies and profile descriptions. Their choice of words, topics, emotional expression and communication style may provide signals that relate to personality. Research has shown that digital footprints can be used to estimate Big Five traits, although prediction strength differs across traits and datasets (Azucar et al., 2018; Farnadi et al., 2016; Tadesse et al., 2018).

The use of online text does not remove the need for careful assessment. A short set of comments may not represent a person's usual communication pattern. Text may be influenced by topic, audience, time, humour or platform conventions. For this reason, personality prediction from text should be treated as an estimate generated from available data, not as a clinical judgement or a basis for an automated decision affecting an individual.

Natural language processing has improved the way written content is represented for machine learning. Earlier approaches often depended on word counts, manually designed

linguistic features or static word vectors. BERT provides contextual representations by considering the words surrounding a token. This helps the model represent meaning that changes with context and has made transformer-based models useful for several text analysis tasks (Devlin et al., 2019).

In personality prediction, however, processing every available comment with a transformer can increase computational cost. Large collections of text may also include repeated, very short or weakly informative comments. The present study therefore includes a Q-learning selector that can learn a select-or-skip policy from the training comments. The policy is designed to favour informative and non-redundant comments within a selection budget. The selected comments are then embedded with BERT.

Another challenge is the limited number of labelled examples in many personality datasets. The training data may not contain sufficient examples in all trait ranges. The current implementation addresses this problem by using a generative adversarial network to create synthetic embedding and OCEAN-score pairs from training data. The augmentation process is focused on underrepresented trait bands and synthetic observations are given a lower training weight than real observations. This approach is evaluated through comparison with conditions that do not use augmentation.

Interpretability is also relevant. A prediction system should make it possible to inspect the procedure, the data split and the performance record behind its output. The regression arm of the experiment uses Lasso or ElasticNet regularization because these methods can reduce the influence of less useful embedding dimensions and retain a more compact model. The LSTM arm provides a sequence-based comparison, but its results are considered alongside the simpler regression baseline rather than assumed to be superior in advance.

The study is therefore motivated by the need to combine practical text processing with a controlled comparison of models. It examines whether contextual text embeddings, Q-learning selection and training-only augmentation can support Big Five prediction from online comments. It also recognizes that performance must be assessed on data that were not used to fit the models.

1.2 Problem Statement

Text-based personality prediction has attracted attention because online platforms provide a large amount of user-generated content. Nevertheless, several problems remain. First, processing complete comment histories with transformer models can be expensive, especially when users contribute many comments. A system that treats every comment as equally useful may spend computational resources on short, duplicated or low-information text without improving the prediction.

Second, personality-labelled text data are not always evenly distributed across trait values. Some trait bands may be represented by fewer examples than others. When models are trained on such data, they may favour common patterns and perform poorly for less represented groups. The shortage of labelled data also makes it difficult to establish whether apparent performance gains are due to useful learning or to overfitting.

Third, it is difficult to compare different prediction methods fairly when the models, text selection procedures and augmentation strategies are changed at the same time. A result from one model cannot show whether an observed improvement came from the model architecture, the selected comments or the augmented data. A structured experiment is needed to isolate these factors.

Fourth, a single validation result is not enough to show that a personality prediction model will generalize. Thresholds selected from the test data can make reported classification results optimistic. Similarly, using comments from the same author in both training and evaluation can make a model appear stronger than it is. The data must therefore be separated by author into training, validation and test sets. Training data should be used for fitting, validation data for model decisions such as threshold selection, and test data for final reporting.

Finally, personality inference is sensitive. Predictions should not be presented as diagnoses or used as a substitute for professional assessment. A research system must retain information about the experimental condition, the data source and the evaluation procedure so that its outputs can be reviewed and interpreted within their limits.

This study addresses these problems by developing a Django-based personality prediction platform and conducting a controlled PANDORA experiment. The experiment compares Lasso or ElasticNet regression and LSTM models across baseline and Q-learning

comment selection, with and without GAN-based augmentation. The evaluation preserves author-level training, validation and test partitions and records regression and classification measures for the experimental conditions.

1.3 Aim and Objectives

The aim of this study is to develop and evaluate a web-based machine learning system for predicting the Big Five personality traits from English-language social media comments using contextual text representations and a controlled comparison of model conditions.

Objectives

1. To prepare personality-labelled social media comments for machine learning and preserve separate author-level training, validation and test datasets.
2. To use a Q-learning method to select informative and non-redundant comments before BERT embedding extraction, and compare it with a baseline comment-selection method.
3. To extract 768-dimensional contextual BERT embeddings from selected comments and construct both pooled features and ordered embedding sequences for prediction.
4. To apply GAN-based training-data augmentation to generate paired embedding and OCEAN-score samples for underrepresented trait bands, and compare augmented conditions with non-augmented conditions.
5. To compare Lasso or ElasticNet regression with a stacked bidirectional LSTM for predicting the five Big Five traits.
6. To evaluate the eight experimental conditions using held-out data and report appropriate regression, classification, threshold and selection-efficiency measures.

1.4 Justification Based on Existing Solutions

Existing studies show that social media text can support personality prediction, but they often use one modelling approach or one type of representation. BERT-based approaches improve text representation because they consider context, while classical regression methods remain useful where simpler and more inspectable models are required (Devlin et al., 2019; Christian et al., 2021). The present study uses both perspectives by retaining a sparse regression baseline and comparing it with a sequence model.

A major difficulty with transformer-based text analysis is that the cost of embedding comments increases with the amount of text processed. Selecting a small but useful subset of comments may reduce unnecessary embedding work. Q-learning is relevant to this problem because it provides a framework for learning actions from reward. In this implementation, the agent learns from text informativeness and redundancy measures while operating under a comment-selection budget. Its contribution is tested against a baseline that selects comments without the learned policy (Sutton & Barto, 2018).

Data augmentation has also been used where real training observations are limited. Standard augmentation techniques may be inadequate for a personality prediction task if a synthetic embedding is not paired with a plausible target score. The implemented GAN therefore generates both an embedding and its associated OCEAN-score vector from a shared generator. The discriminator evaluates the combined pair. This allows the augmented examples to remain usable for supervised training, although their value must still be verified by held-out results rather than assumed (Goodfellow et al., 2014; Mirza & Osindero, 2014).

Lasso and ElasticNet regression are included because regularization can reduce the influence of redundant features and provide an understandable baseline for high-dimensional embeddings (Tibshirani, 1996; Friedman et al., 2010). The LSTM model is included because the order and variation of comments may contain information that is lost when embeddings are averaged. Comparing both models under the same datasets and selection conditions gives a more useful basis for judging their behaviour.

The use of distinct training, validation and test datasets further justifies the proposed approach. The training set is used to fit the Q-learning selector, GAN and prediction models. The validation set is used to choose decision thresholds. The test set is reserved for final reporting. This procedure reduces the risk of choosing model settings based on the same data used to claim final performance.

1.5 Scope of the Study

This study is limited to the prediction of the five Big Five personality traits from English-language online comments. The main experiment uses the PANDORA dataset and its supplied personality labels. The unit of analysis is an author represented by a collection of cleaned comments. The study does not use profile photographs, video, audio, private messages or other multimodal content.

The experiment uses separate PANDORA training, validation and test datasets. Authors with sufficient usable comments are represented in the model pipeline. The latest experimental configuration samples a defined number of eligible authors, applies a comment-selection budget and uses BERT embeddings of selected text. The data are split at author level so that comments from one author do not appear in more than one experimental partition.

The machine learning scope covers eight conditions formed from two prediction models, two comment-selection approaches and two augmentation settings. The regression arm uses Lasso or ElasticNet models on pooled BERT embeddings. The sequence arm uses a stacked bidirectional LSTM on ordered comment embeddings. The selection comparison is between a baseline method and a Q-learning method. The augmentation comparison is between original training observations and a training set expanded with GAN-generated paired samples.

The system is implemented in Django and stores experiment records, condition results, dataset allocations, threshold results and prediction profiles. It can also provide a researcher interface for initiating an experiment and reviewing stored results. The public demonstration interface is outside the main PANDORA evaluation and should not be treated as a substitute for the trained experimental pipeline.

The application domains considered are education, health and wellbeing, employment and responsible artificial intelligence. Within this study, they are presented as possible areas for future research and careful decision support. The system is not designed for clinical diagnosis, recruitment screening, disciplinary action or any fully automated decision about an individual.

1.6 Limitations of the Study

The study has several limitations. First, the data consist of English-language online comments from a particular dataset. Language use on other platforms, in other cultures or in other languages may differ. The findings should therefore not be generalized automatically beyond the population and data source used in the experiment.

Second, online comments provide only a partial view of an author. Topic, community norms, humour and temporary circumstances can affect what a person writes. Even where a model identifies statistical associations, its output remains an estimate of a questionnaire-linked personality score rather than a complete account of the author.

Third, the dataset is modest for deep learning and some trait bands are less common than others. GAN augmentation can increase the number of training examples, but generated examples are not new human observations. The augmentation may help some model conditions and may not help others. Its effect must be interpreted from the controlled comparisons.

Fourth, the present Q-learning reward is based on text informativeness, novelty and selection cost. It is not a direct reward from improvement in personality prediction accuracy. The selector may reduce the number of comments that require BERT embedding, but its effect on final performance needs to be judged from the matched baseline comparison.

Fifth, the study evaluates the experimental pipeline offline. It does not establish long-term stability, test-retest reliability or performance in a live deployment. The public demo interface is heuristic and does not call the trained PANDORA model for every request. Its output should therefore be treated as a demonstration feature rather than experimental evidence.

Finally, the use of pre-trained language models and personality-labelled online text creates ethical concerns. Bias in the source data or the language model may affect the predictions. The system should be used only for research and educational purposes with clear explanation of its uncertainty, data limitations and appropriate safeguards.

1.7 Definition of Terms

1. Agreeableness: A Big Five trait associated with compassion, cooperation, trust and concern for others.
2. BERT: Bidirectional Encoder Representations from Transformers, a pre-trained language model used to produce contextual text embeddings.
3. Big Five Personality Traits: The five dimensions of Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism used to describe personality.
4. Embedding: A numerical vector representation of text used as input to a machine learning model.
5. ElasticNet Regression: A regularized regression method that combines L1 and L2 penalties to reduce overfitting in high-dimensional data.
6. Generative Adversarial Network: A model containing a generator and discriminator that is trained to produce synthetic examples resembling the training data.
7. Lasso Regression: A regression method that uses L1 regularization to reduce the influence of less useful features.
8. Long Short-Term Memory Model: A recurrent neural-network model used in this study to process sequences of BERT comment embeddings.
9. PANDORA Dataset: A personality dataset of online comments and associated Big Five personality scores used for the main experiment.
10. Q-Learning: A reinforcement-learning method that learns expected rewards for actions in a given state. In this study, it is used to select or skip comments before embedding.
11. Validation Set: A dataset used during model development for selecting settings such as classification thresholds, separate from the final test set.